Structure at 2.8 Å resolution of F_1-ATPase from bovine heart mitochondria

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In the crystal structure of bovine mitochondrial F_1-ATPase determined at 2.8 Å resolution, the three catalytic β-subunits differ in conformation and in the bound nucleotide. The structure supports a catalytic mechanism in intact ATP synthase in which the three catalytic subunits are in different states of the catalytic cycle at any instant. Interconversion of the states may be achieved by rotation of the $\alpha_3\beta_3$ subassembly relative to an α-helical domain of the γ-subunit.

ADENOSINE triphosphate synthase (ATP synthase, F_1F_o-ATPase) is the central enzyme in energy conversion in mitochondria, chloroplasts and bacteria (reviewed in refs 1–6). It uses a protonmotive force, generated across the membrane by electron flow, to drive the synthesis of ATP from ADP and inorganic phosphate[7]. The multisubunit assembly consists of a globular domain, F_1, and an intrinsic membrane domain, F_o, linked by a slender stalk about 45 Å long. The F_1 domain is an approximate sphere 90–100 Å in diameter, and contains the catalytic binding sites for the substrates ADP and inorganic phosphate. Energy released by proton flux through F_o is relayed to the catalytic sites in the F_1 domain, probably by conformational changes through the stalk. About three protons flow through the membrane per ATP synthesized.

Disruption of the stalk releases the water soluble enzyme, F_1-ATPase, a complex of five different proteins with the stoichiometry $3\alpha:3\beta:1\gamma:1\delta:1\varepsilon$. In bovine mitochondria they contain 510, 482, 272, 146 and 50 amino acids, respectively[8], giving a relative molecular mass of 371,000. The sequences of the α- and β-subunits are homologous (20% identical), including the P-loop nucleotide-binding motif[9]. The catalytic sites are in the β-subunits, whereas the function of nucleotides in the α-subunits, which do not exchange during catalysis, is obscure. According to a binding change mechanism of ATP synthesis[1,10], the structures of the three catalytic sites are always different, but each passes through a cycle of 'open', 'loose' and 'tight' states. The mechanism suggests that F_1-ATPase is an inherently asymmetric assembly, as clearly indicated by subunit stoichiometry, electron microscopy[11,12] and low-resolution X-ray crystallography[13]. However, in one model proposed for rat F_1-ATPase based on analysis of a crystal form that implies three-fold symmetry, the three different α- and β-subunits are indistinguishable[14].

To obtain a fuller understanding of the mechanism of F_1 and of the non-equivalence of the catalytic sites, we have determined the structure of F_1-ATPase from bovine heart mitochondria by X-ray diffraction at 2.8 Å resolution. Bovine F_1-ATPase is the largest asymmetric structure solved at atomic resolution so far. The α- and β-subunits, which have a similar fold, are arranged alternately like the segments of an orange around a central α-helical domain containing both the N- and C-terminal regions of the γ-subunit. As the three β-subunits contain different nucleotides and have different conformations, the structure as found in the crystal is compatible with one of the states to be expected in the cyclical binding change mechanism. It is likely that the central helical domain is part of a mechanism for transferring energy from the F_o domain into the catalytic sites, and that in the process the 3α–3β assembly (referred to below as $\alpha_3\beta_3$) may rotate about the central α-helical domain of the γ-subunit.

Structure determination

Crystallization of bovine F_1 and data collection procedures have been described[13,15]. The quality of data and initial phases determined by single isomorphous replacement and anomalous scattering are summarized in Table 1, and a representative sample of the electron density map is shown in Fig. 1. The model (Fig.

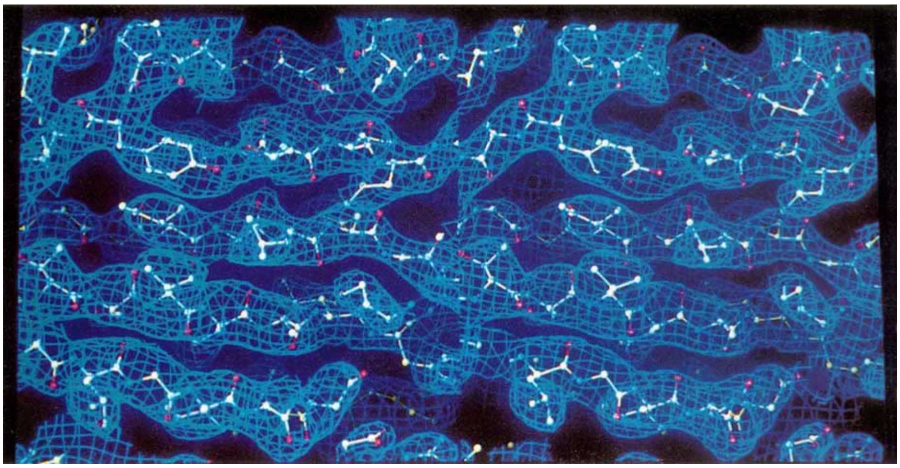


FIG. 1 Stereo illustration of part of the 3.2 Å-resolution electron density map used to build the initial model of bovine F_1-ATPase, with the refined model coordinates superimposed. The region shown corresponds to part of the β-sheet in the nucleotide-binding domain of subunit β_{DP}. The contour level is 1.2 times the r.m.s. density of the map.

TABLE 1 Crystallographic data

	Data set 1		Data set 2		Data set 3	
	Native	0.2 mM methylmercury	Native	1 mM methylmercury	Native	0.2 mM methylmercury
Resolution	2.85 Å	3.2 Å	3.5 Å	3.5 Å	3.5 Å	3.5 Å
R_{merge}*	0.098 (0.19)	0.077 (0.145)	0.059 (0.106)	0.067 (0.111)	0.114 (0.255)	0.086 (0.210)
Completeness	97.5% (92%)	96.3% (95.6%)	97.7% (87.7%)	97.4% (90.4%)	97.5% (99.8%)	93.5% (91.5%)
Multiplicity	5.4 (2.9)	3.7 (3.5)	3.2 (2.8)	7.5 (6.4)	8.9 (7.8)	2.4 (2.0)
No. of crystals	17	6	1	6	12	3
Rejected outliers†	0.37%	0.02%	0.56%	0.4%	0.3%	0.4%
Mean $\langle F/\sigma(F)\rangle$	30.9 (7.8)	28.8 (11.8)	46.9 (24.8)	53.8 (30.0)	32.6 (15.1)	19.2 (8.9)
R_{deriv}‡		0.114 (0.164)		0.120 (0.132)		0.170 (0.215)
No. of mercury sites		4		9		4
Total occupancy§		2.4		3.0		2.4
R_{cullis}‖		0.60 (0.80)		0.56 (0.71)		0.72 (0.87)
Phasing power¶		1.37 (0.95)		2.0 (1.6)		1.08 (0.78)

Diffraction data. As native crystals were sometimes non-isomorphous, every time a derivative data set was collected, a corresponding native data set was collected too. Data sets 1 and 2 were collected on stations 9.6 and 9.5 respectively at the Synchrotron Radiation Source, Daresbury, UK. Data set 3 was collected using an Elliot GX11 generator with a 100 μ focal spot, collimated by a Supper double-mirror system. Data sets were recorded on MAR Research image-plate detectors, with a diameter of 300 mm for data sets 1 and 3, or 180 mm for data set 2. Data were processed with MOSFLM[43] and programs from the CCP4 suite[44]. Values given in parenthesis are for the highest resolution bin. The R_{deriv} between native data sets 1 and 3 was 0.074, that between data sets 1 and 2 was 0.157, and that between data sets 2 and 3 was 0.162, indicating that data set 2 was not isomorphous with data sets 1 and 3, and that data sets 1 and 3 were not isomorphous to a lesser degree. Data from derivatized crystals were merged only if refinement of the heavy-atom parameters of the individual crystals indicated that this was allowed; otherwise they were treated separately. The phase-probability distributions of different data sets were combined, but only the structure-factor amplitudes of the high-resolution data set were used in the structure determination. **Solvent density modification.** Phases to 3.2 Å were improved by a new type of solvent density modification, to be described elsewhere. The solvent region (50% of the crystal volume) was determined from the local variation in electron density within a sphere of radius 3.7 Å. Features in the solvent region were inverted with respect to the mean solvent density. The phases of the resulting map were combined with the original phase probability distribution using σ_A weighting[45], and a new map was calculated. Iteration of this process decreased the mean phase error from 70 to 40°. Phase improvement was monitored by measuring the reduction in variation of the electron density within an unmodified spherical patch of solvent with a radius of 8 Å. **Model building and refinement.** The atomic model was built using program O (ref. 46). The few ambiguities in following the path of the polypeptide backbone of the α- and β-subunits were resolved by comparing the electron density of different copies of these subunits. Conventional refinement using XPLOR-3.1 (ref. 47) resulted in a model with an R-factor of 19.4% for 95% of the data between 6.0 and 2.86 Å. The free R-factor[48] for the remaining 5% of the data within this resolution range was 28.5%. The r.m.s. deviations from standard values of bond lengths and angles are 0.016 Å and 2.1°.

* $R_{merge} = \sum\sum_i |I(h) - I(h)_i| / \sum\sum_i I(h)_i$, where $I(h)$ is the mean intensity after rejections. The contributing reflections were weighted by their standard deviations, which were determined by adjusting the measured standard deviation to reflect the observed differences between symmetry related reflections.

† Reflections with intensities differing more than 3.5 $\sigma(I)$ from the weighted mean were rejected.

‡ $R_{deriv} = \sum ||F_{PH}| - |F_P|| / \sum |F_P|$, where F_{PH} is the structure factor of the derivative and F_P is that of the native data.

§ The total refined occupancy of the sites, normalized to the site with the highest occupancy after setting this to 1.

‖ $R_{cullis} = \sum ||F_{PH} - F_P| - |F_{H(calc)}|| / \sum ||F_{PH} - F_P|$; F_P and F_{PH} are defined above, $F_{H(calc)}$ is the calculated heavy-atom structure factor. The summation is over centric reflections only.

¶ R.m.s. isomorphous difference divided by the r.m.s. residual lack of closure.

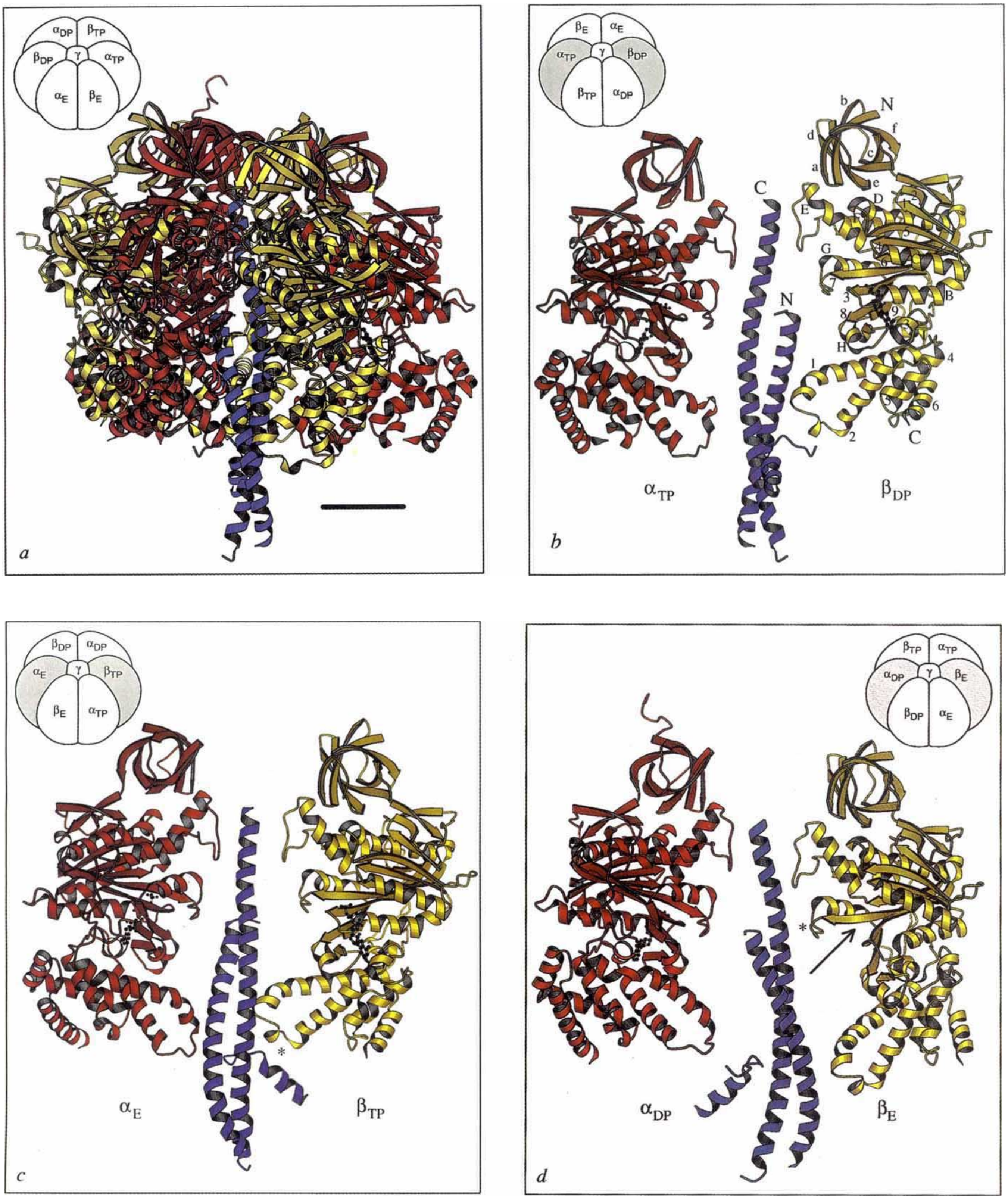
2) contains 2,983 amino acids in the following sequences: α_{DP} 19–510, α_{TP} 25–510, α_E 24–510, 9–474 in all three β-subunits, γ1–45, γ73–90 and γ209–272 (see Fig. 2 legend for subunit nomenclature). Four AMP-PNP molecules and one ADP are also present (where AMP-PNP is 5′-adenylyl-imidodiphosphate). No water molecules were modelled into the structure. The δ- and ε-subunits and 145 residues of the γ-subunit were not located in the electron density map.

It was possible to solve the structure with data from a single heavy-atom derivative as a result of recent advances in detector technology. Apart from the new solvent density modification method (Table 1), standard techniques were used.

Architecture of F_1-ATPase

F_1 is a flattened sphere 80 Å high and 100 Å across. The three α- and three β-subunits are arranged alternately like the segments of an orange around a central α-helix 90 Å long formed by the C-terminal amino acids γ209–272 (Fig. 2). The C-terminal residue of this helix emerges into a dimple 15 Å deep in the surface at the top of the assembly. The lower half of the helix forms a left-handed antiparallel coiled coil with a second α-helix composed of amino acids γ1–45. This helical structure protrudes about 30 Å from the bottom of the main body in a stem, almost certainly a vestige of the 45 Å stalk in ATP synthase. An increased local variation of the electron density suggests that the δ- and ε-subunits are probably in the stem region, but are not well ordered. However, it cannot be excluded that the δ- and ε-subunits are sub-stoichiometric in the crystals. The γ-subunit passes through a large internal cavity connected to the outside by crevices between neighbouring α- and β-subunits. In the low-resolution structure, this cavity was interpreted to be a pit[13]. The C-terminal helix of the γ-subunit is twisted (Fig. 2). At the top

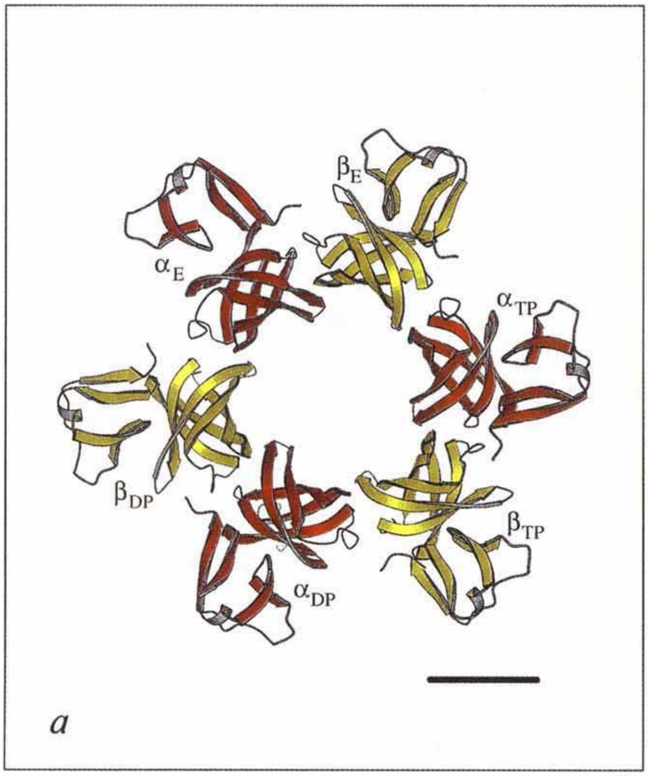
FIG. 2 The three-dimensional structure of bovine F_1-ATPase. In this figure and Fig. 3, parts *a–c* were prepared with MOLSCRIPT[49]. The α-, β- and γ-subunits are red, yellow and blue, respectively, and nucleotides are black, in a 'ball-and-stick' representation. The axis of pseudo-symmetry is vertical. AMP-PNP is bound to the three α-subunits, and to the β-subunit defined as β_{TP}. Subunit β_{DP} has bound ADP and subunit β_E has no associated nucleotide. The relationships of the various α- and β-subunits to each other is summarized in the top left or right corner (*a–d*). In *b–d*, the shaded part of the diagram indicates which subunits are depicted. Subunit α_{TP} contributes to the nucleotide-binding site of β_{TP}, and similarly for α_{DP} and α_E. Subunits α and γ are numbered from 1–510 and 1–272, respectively. By convention, the fifth amino acid (serine) in subunit β is residue 1 and the first four amino acids (AlaAlaGlnAla) are referred to as residues −1 to −4. The C-terminal amino acid is residue 478. *a*, A view of the entire F_1 particle in which subunits α_E and β_E point towards the viewer, revealing the antiparallel coiled coil of the N- and C-terminal helices of the γ-subunit through the open interface between them. The bar is 20 Å long. *b*, Subunits α_{TP}, γ and β_{DP}. The N and C termini of the β- and γ-subunits are shown. The

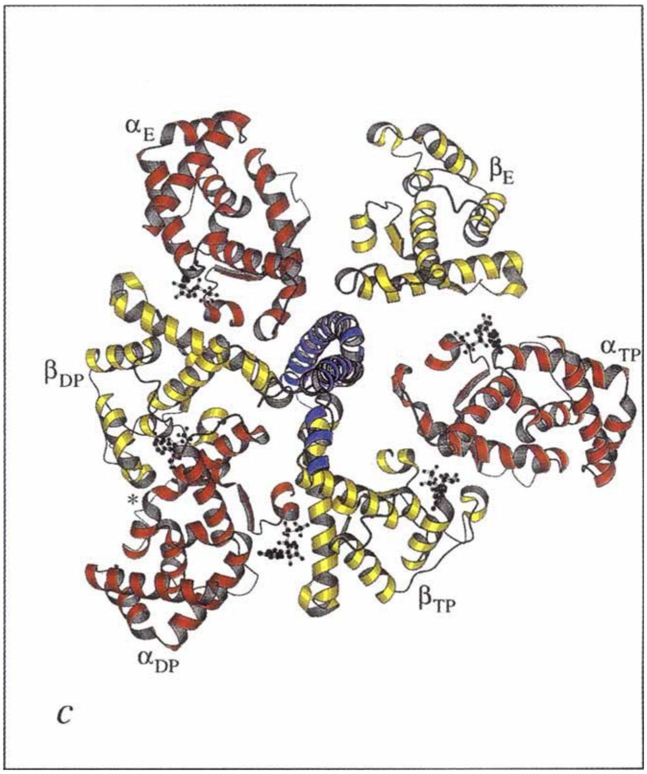


β-sheets in the N-terminal β-barrel are labelled a–f, consecutively. The order of helices (A–I) and the β-sheets (1–9) in the nucleotide-binding domain is as follows: 1, 2, A (not visible, hidden by helix 3 from the C-terminal domain), 3, B, 4, C, 5, D, 6, E, F, 7, G, 8, H, 9, I. The helices in the C-terminal domain are labelled 1–6, consecutively. The α-subunits have the same fold, with a minor deviation in the C-terminal domain, where the equivalent of helix 3 in the β-subunit is missing, and where two more helices are present after the equivalent of helix 6. *c*, Subunits α_E, γ and β_{TP}. The asterisk indicates a 'catch' interaction of the loop containing the DELSEED sequence and the γ-subunit. *d*, Subunits α_{DP}, γ, and β_E. The arrow indicates the disruption of the β-sheet in the nucleotide-binding domain. β_E-Asp 316, β_E-Thr 318 and β_E-Asp 319 in a loop of the nucleotide-binding domain hydrogen-bonds with residues γ-Arg 254 and γ-Gln 255 from the C-terminal helix, 6 Å below the hydrophobic sleeve. The asterisk indicates the loop that makes a 'catch' with the C-terminal part of the γ-subunit. *e*, *f*, Stereo α-carbon backbone drawings of subunits α_{TP} and β_{DP}, respectively, in the same orientations as in the β-subunit in *b*.

of the assembly, its axis is coincident with the pseudo-threefold axis relating the α- and β-subunits, but at the lower end of the stem it is displaced from the axis by 7 Å. A third segment of the γ-subunit (γ73 90) forms a small α-helix and loop in the region of the stem (Fig. 2).

The folds of the α- and β-subunits are almost identical, so that elements of three- and six-fold symmetry are present in the structure. Each subunit consists of an N-terminal six-stranded β-barrel (α19 95, β9 82), a central α β domain containing the nucleotide-binding site (α96 379, β83 363), and a C-terminal bundle (α380 510, β364 474) of 7 and 6 helices, respectively, in the α- and β-subunits (Fig. 2). In each subunit, the β-barrel and nucleotide-binding domains are separated, whereas the nucleotide-binding domain and the helical bundle interdigitate to a minor extent. The extent of contact between neighbouring α- and β-subunits varies (Fig. 3). The β-barrel domains crown the particle with a continuous 24-stranded β-sheet (Fig. 3*a*), whereas the shortest distance between side chains of some of the helical bundles (Fig. 3*c*) is 10 Å. Between 18 and 24 amino acids at the N termini of α-subunits and amino acids $\beta(-4$ to $8)$ are disordered. In F_1-ATPase and in ATP synthase, these regions are protease-sensitive[8,16].





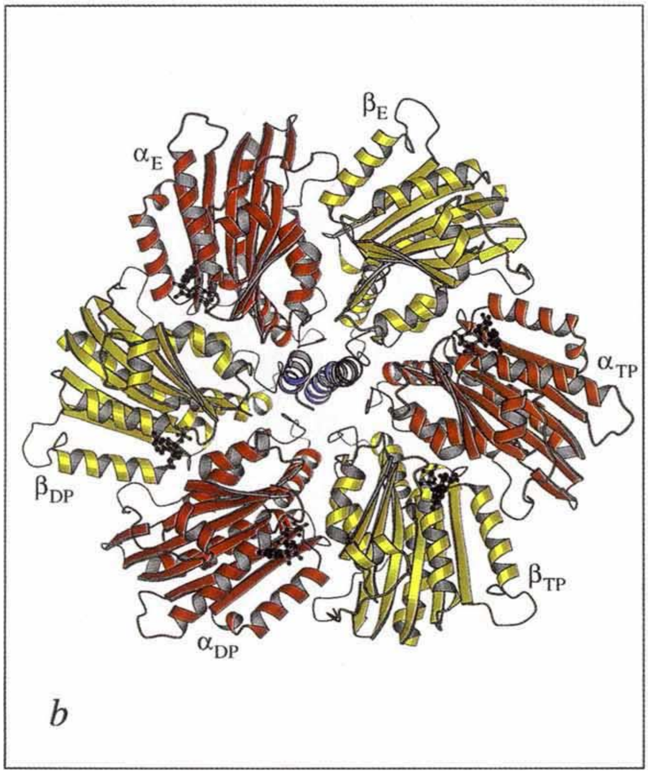
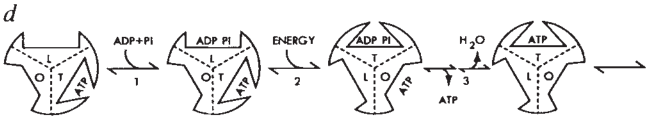


FIG. 3 Asymmetrical features in bovine F_1-ATPase and the binding change mechanism; *a–c* are viewed from the membrane side, the three views being drawn to the same scale and in the same orientation (rotated by 90 relative to Fig. 2*a*). *a*, The N-terminal domains and part of the nucleotide-binding domains of α- and β-subunits. The N-terminal domains make a 24-stranded β-barrel with the six a-strands hydrogen-bonded to d-strands of the neighbouring subunit (Fig. 2*b*). Scale bar, 20 Å. *b*, The nucleotide-binding domains. The β-sheet of subunit β_E is disrupted close to the γ-subunit. *c*, The C-terminal domains. The deviations from perfect symmetry (most apparent in this view) result in the characteristic features of the individual nucleotide-binding sites. *d*, An energy-dependent binding change mechanism of ATP synthesis. The catalytic sites in β-subunits interact and interconvert between three forms: O, open site with very low affinity for ligands and catalytically inactive; L, loose site, loosely binding ligands and catalytically inactive; T, tight site, tightly binding ligands and catalytically active. The proton-induced conformational changes convert a T-site with bound ATP to an O-site, releasing the bound nucleotide. Concomitantly, an L-site with loosely bound ADP and phosphate is converted to a T-site, where the substrates are tightly bound and ATP forms. Fresh substrates bind to an O-site, converting it to an L-site, and so on. The energy-requiring steps are substrate binding and product release. One third of the catalytic cycle is shown. (Reproduced with permission from ref. 1).

The nucleotide-binding domain contains a nine-stranded β-sheet with nine associated α-helices. In both α- and β-subunits, β-strands 3–9 and helices A–G have the same topology as β strands 1–7 and associated helices in RecA[17]. Within this region, between 113 and 145 α-carbons superimpose on RecA with an rms separation of 1.9–2.0 Å (depending on the subunit). The nucleotide pyrophosphate group binds in the same way in both structures, but the orientation of the adenosine moiety is very different. In adenylate kinase[18], often used as a model for F_1-ATPase, the corresponding β-sheet has five strands with different connectivity.

d



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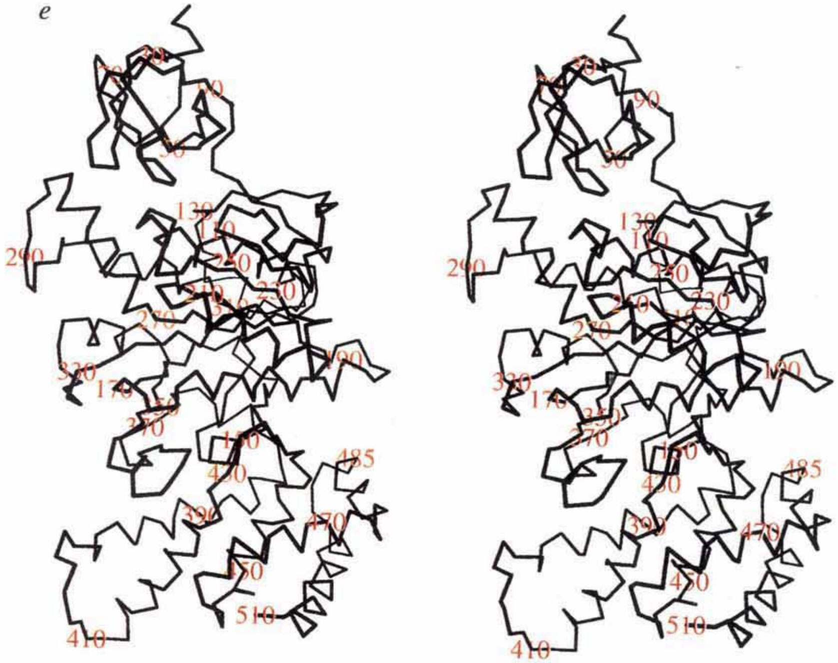

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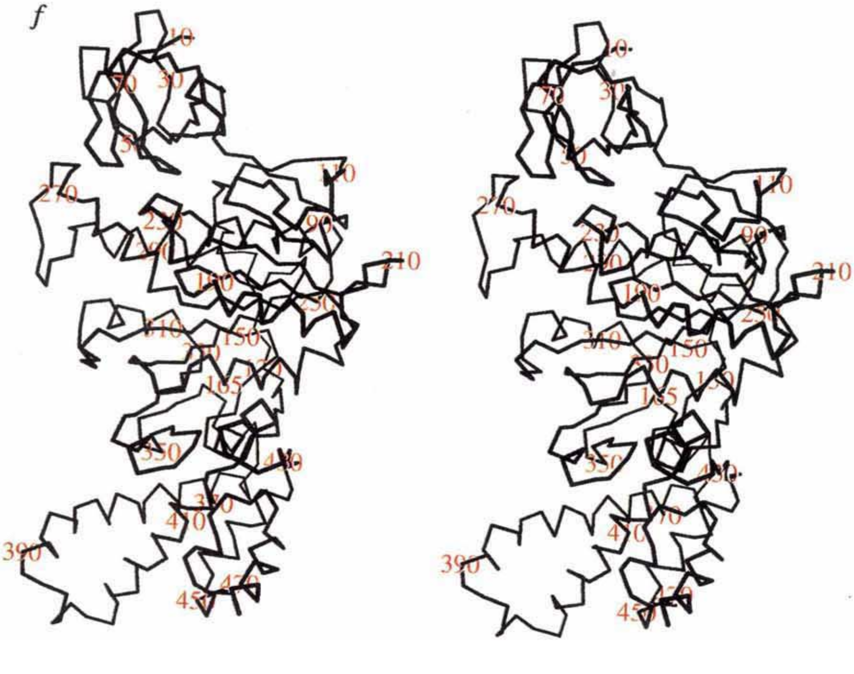


Nucleotide-binding sites

The nucleotide-binding sites are at the interfaces between α- and β-subunits. The catalytic sites are predominantly in β-subunits, with some contributions from side chains in the α-subunits, and conversely with the non-catalytic sites. Most notably, β-Tyr 368, a site of reaction of both photoactivated 2-azido-ADP[19] and of the nucleotide analogue 5′-p-fluorosulphonylbenzoyl-adenosine[20], protrudes into the α-subunit nucleotide site (Fig. 4). The average distance between adjacent catalytic and non-catalytic sites is 27 Å (between β-phosphates). In both α- and β-subunits, one face of the ribose and phosphate moieties is accessible to external solvent via a conical tunnel. In the crystals, the subunit denoted β_{TP} and each of the three α-subunits bind one molecule of the nucleotide triphosphate analogue AMP-PNP, whereas the diphosphate ADP is bound to β_{DP}. In subunit β_E, the nucleotide-binding site is empty and distorted, despite the presence of 250 μM AMP-PNP and 5 μM ADP in the solvent[15]. Inorganic phosphate is not present in the ADP site, and so the structure is probably that of the ADP-inhibited state[10]. The three α-subunits are defined in relation to β_{TP}, β_{DP} and β_E. Subunit α_{TP} forms the interface with β_{TP} that contains the β_{TP} nucleotide site, and similarly for α_{DP} and α_E (Fig. 2).

The amino acids involved in nucleotide binding are summarized in Fig. 4. Those that define the adenine- and ribose-binding pocket in the β-subunits include β-Tyr 345, which forms a crosslink with photoactivated 2-azido-ATP[21]. In contrast, β-Tyr 311, which reacts with photoactivated 8-azido-ATP[22,23], is 5.5 Å from the terminal phosphate at the opposite end of the binding site to the adenine pocket. The reaction appears to require non-productive binding of the nucleotide. The most important phosphate-binding amino acids are in the P-loop, which can be seen between strand 3 and helix B in subunit β_{DP} in Fig. 2*b*. This loop contains the conserved nucleotide-binding motif, GXXXXGKT/S, first recognized in the sequences of the α- and β-subunits of F_1-ATPase[9]. The motif has been used to predict the nucleotide-binding sites of many other proteins. The crystal structures of p21^{ras} (ref. 24), adenylate kinase[25], RecA[26], elongation factor Tu[27,28] and transducin-α[29], for example, show that the motif binds the di- or triphosphate moiety, and this is also true in the α- and β-subunits of F_1-ATPase (Fig. 4).

The crystal structure provides new details of the chemical mechanism of ATP synthesis. In β_{TP}, at a distance of 4.4 Å from the γ-phosphate of AMP-PNP, there is clear density for a bound water molecule, hydrogen-bonded to the carboxylate of β-Glu 188. This residue is appropriately positioned to activate the water molecule, promoting an in-line nucleophilic attack on the terminal phosphate. The guanidinium of α-Arg 373 could help to stabilize the negative charge that develops on the terminal phosphate in a pentacoordinate transition state. A similar arrangement of a glutamic acid and an arginine side chain is found in the active site of transducin-α complexed with GTP-γS[29]. The structure also indicates why the α-subunits

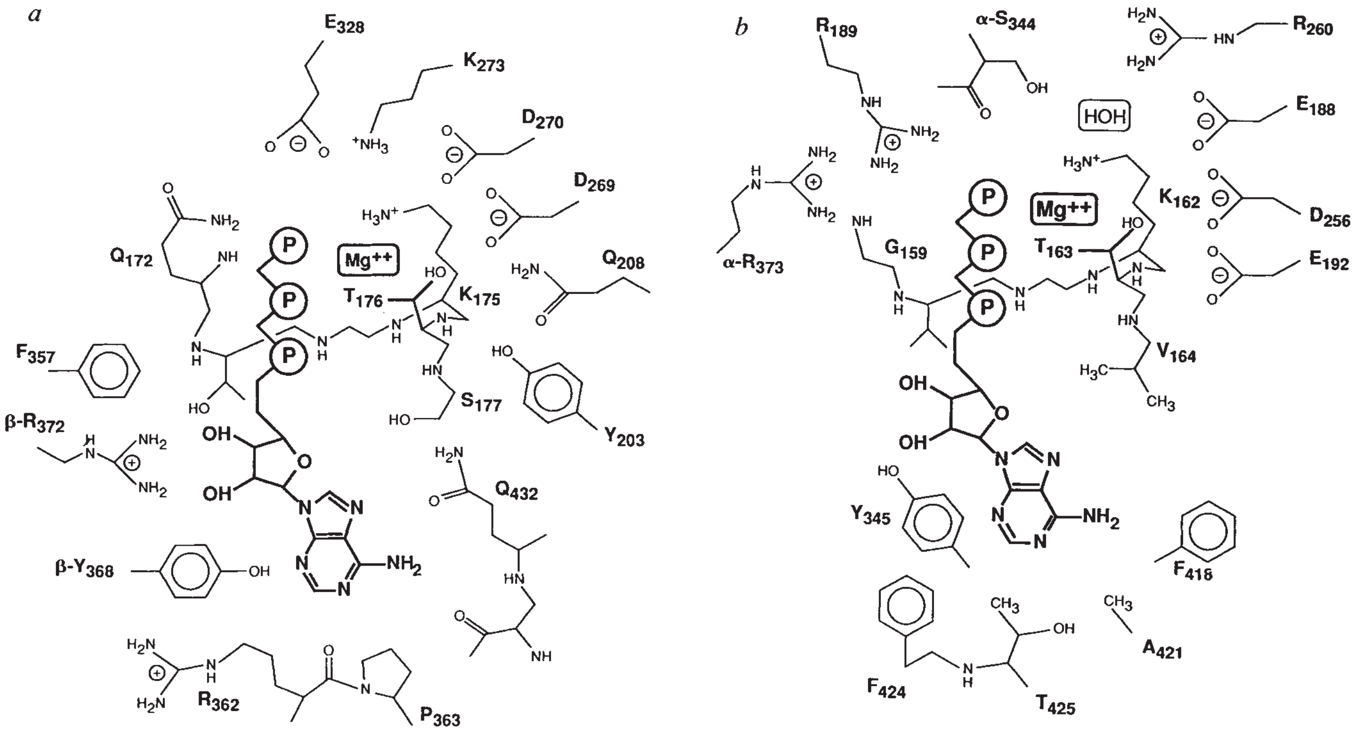


FIG. 4 The nucleotide-binding sites in the α- and β-subunits of bovine F_1-ATPase. *a*, Subunit α_E. All of the amino acids forming the pocket are in the α-subunit, except for the neighbouring β_{DP}-Arg 372 and β_{DP}-Tyr 368. Residues α172–177 are part of the P-loop. The nucleotide-binding sites of α_{DP} and α_{TP} are very similar, except that β-Arg 372 does not contribute to either binding site, and β-Tyr 368 only contributes to nucleotide binding in α_{DP}. *b*, Subunit β_{TP}. Except for α_{TP}-Arg 373 and the main-chain carbonyl and the side chain of α_{TP}-Ser 344, all of the amino acids are in β_{TP}. Residues 159–164 are part of the P-loop. In β_{DP}, there is density for an ordered water molecule replacing the terminal phosphate, and Arg 373 moves closer to the diphosphate moiety. Except for β-Arg 372 in the α-subunit nucleotide-binding site, all residues forming the pockets are widely conserved. The ligands to magnesium identified at this stage are as follows: in the α-subunits, O_γ-Thr 176, two oxygens from the terminal phosphate and one from the β-phosphate; in β_{TP}, O_γ-Thr 163 and one oxygen atom from each of the β- and terminal phosphates; in β_{DP}, O_γ-Thr 163 and one oxygen from the β-phosphate. The remaining ligands are presumably water molecules.

are non-catalytic. In the α-subunits, there is no spatial equivalent of the carboxylate of β-Glu 188, which is replaced in α-subunits by Gln 208, with the side chain pointing away from the terminal phosphate.

The binding of adenine by the α-subunit involves several hydrogen bonds, whereas in the β-subunit the adenine is in contact with a hydrophobic interface. Together with the presence of β-Tyr 368 close to the 2-position of the adenine ring in the α-subunit, this may explain the specificity of the α-subunit nucleotide sites for adenine nucleotides, and the ability of the β-subunit nucleotide sites to accommodate GTP and ITP as well as ATP[6,30]. The roles of the non-exchangeable nucleotides in the α-subunits remain obscure.

The site of reaction of the inhibitor dicyclohexylcarbodiimide (β-Glu 199; ref. 31) points into the conical tunnel leading to the nucleotide sites. The presence of the two bulky cyclohexyl moieties in the conical tunnel could impede nucleotide binding and the conversion of the catalytic sites between conformational states.

Asymmetrical features

With the exception of the nucleotide-binding domain of β_E, the domains of the three α- and three β-subunits superimpose with a root-mean-squared separation of less than 1 Å between Cα atoms, but because of differences in the relative orientations of the domains and the interactions with the unique γ-subunit, the overall structure of F_1 is strikingly asymmetric (Fig. 3).

In the β_E nucleotide-binding domain, the interaction between strands 3 and 7 is disrupted (above and below the arrow in Fig. 2*d*). The lower part of this domain and the α-helical domain are rotated by 30° with respect to their positions in β_{TP} and β_{DP}, resulting in movements by up to 20 Å. At the C terminus of helix B following the P-loop (below the arrowhead in Fig. 2*d*), the disruption of the β-sheet is accompanied by relaxation of a few residues into an α-helical conformation. A number of other hydrogen-bonding rearrangements occur throughout the rest of β_E. Also, β_E-Asp 316, β_E-Thr 318, and β_E-Asp 319 in the loop following strand 7 (indicated by the asterisk in Fig. 2*d*) hydrogen-bond with γ-Arg 254 and γ-Gln 255 in the C-terminal helix. Close by, α_E-Asp 333 makes a salt bridge with γ-Arg 252. These H-bonds form a 'catch', and are the major specific interactions between the two long helices of the γ-subunit and the subassembly $\alpha_3\beta_3$.

A second 'catch' is found in the C-terminal domain of subunit β_{TP}, but not in β_{DP} or β_E. Additional hydrogen bonds link γ-Lys 87, γ-Lys 90 and the backbone amide of γ-Ala 80 (all in the short helix, which is inclined at about 45° to the axis of pseudosymmetry) with β_{TP}-Asp 394 and β_{TP}-Glu 398 in a loop between helices 1 and 2. This sequence (DELSEED, β 394–400) is part of the binding site of amphipathic cationic inhibitors[32,33] and possibly of the ATPase inhibitor protein[34] in one or other of the β-subunits.

Asymmetry is also evident in α-subunits. The non-catalytic nucleotide-binding sites in α_{DP} and α_E are very similar, but in α_{TP} the site is more open because the lower half of β_E (which contributes to this binding site) is hinged out. Therefore, β_E-Tyr 368 no longer interacts with the N3 atom of the adenosine. It is likely that α_{TP} binds the exchangeable non-catalytic nucleotide observed in nucleotide-binding studies[10].

Asymmetry is also manifest in the coiled coil of the γ-subunit. Its C terminus (γ265–272) passes through a hydrophobic sleeve

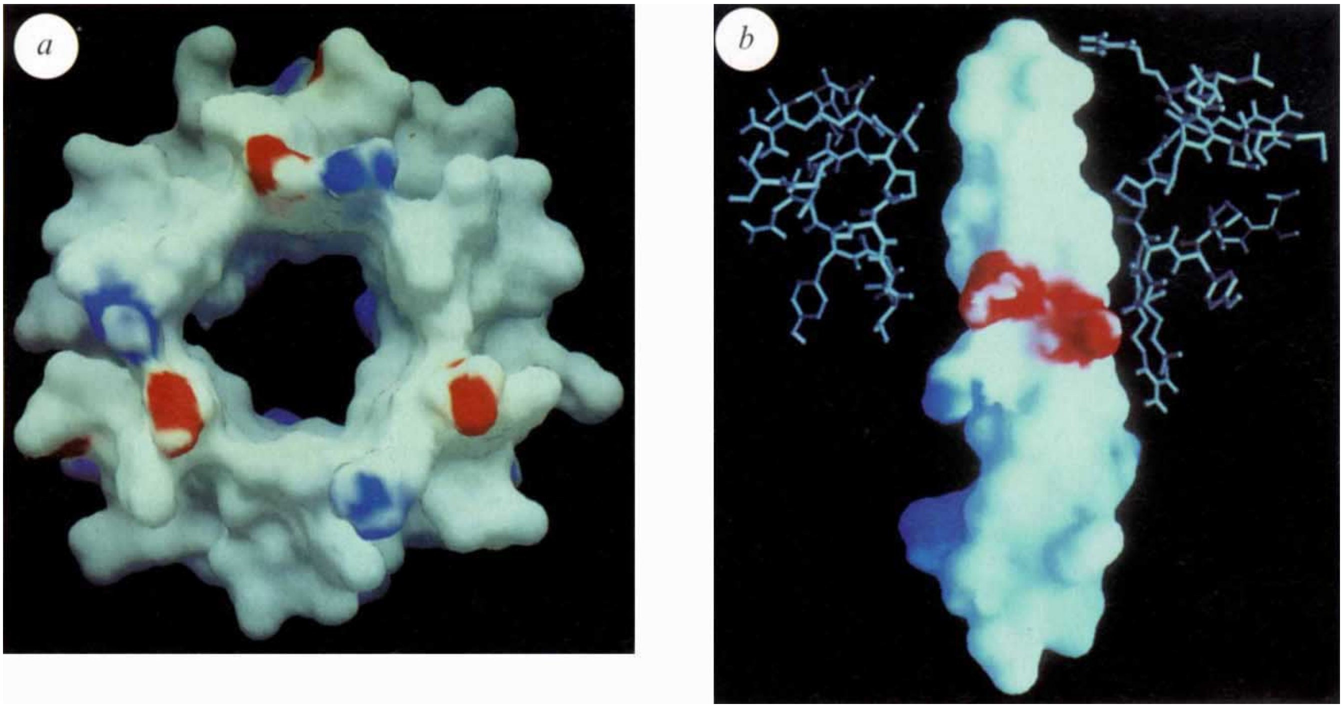
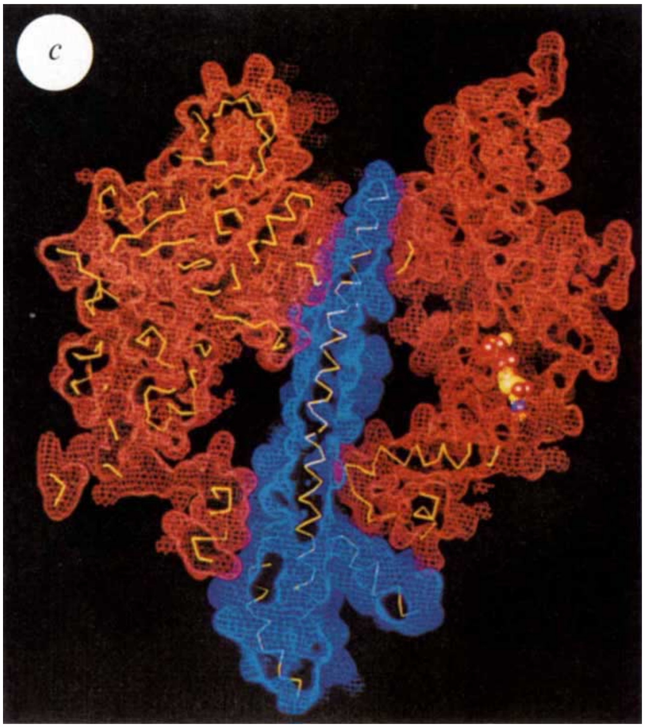


FIG. 5 Features of the structure of F_1-ATPase that are compatible with rotational catalysis. *a*, The electric surface potential of the hydrophobic sleeve surrounding the C-terminal tip of the γ-subunit, viewed from inside the assembly. The sleeve is made from the regions of the 3α and 3β subunits immediately below the crown of β-barrels. The surface was calculated with GRASP[50] from the coordinates of α287–294 and of β274–281. Negative and positive electric potentials of the side chains are red and blue, respectively; white is neutral. *b*, The electric surface potential of the C-terminal helical domain of the γ-subunit (residues γ253–272) shown with a 'ball-and-stick' representation of the amino acids from two diametrically opposed loops making the sleeve in *a*. Because no charged residues are present in the sleeve, the assembly could act as a molecular bearing, lubricated through a hydrophobic interface. *c*, Cross-section of F_1-ATPase showing the central cavity (black) traversed by the γ-subunit. The cross-section through an electron density map was generated from the atomic model and contoured at a level corresponding to the van der Waal's surface. The density for the α- and β-subunits is orange, and that for the γ-subunit is blue. The α-carbon backbones of the α-, β- and γ-subunits (red, yellow and yellow, respectively), and a CPK model of AMP-PNP are presented.



formed by six proline-rich loops (α287–294, β274–281; following helix E in β_{DP} in Fig. 2*b*, and the related loop in α_{TP}). The remaining interactions with the γ-subunit take place through hydrophobic loops, some of them poorly ordered, in the C-terminal domains of the α- and β-subunits, and helix 1 in β_{DP}. In the crystal structure, the γ-subunit seems to prevent β_E from adopting a nucleotide-binding conformation by obstructing the rotation of its lower half towards the central axis. In consequence, the lower halves of β_E and α_{TP} do not interact, and the middle and C-terminal domains of α_{TP} have moved away from the central axis (Fig. 3*c*). Therefore, each β-subunit adopts a different orientation towards the α-subunit that contributes to its active site, and the active site interfaces of β_{DP}, β_{TP} and β_E appear from a structural point of view to be tightly closed, more open, and fully open, respectively. This suggests that β_{DP} contains the non-exchangeable catalytic site[10]. The closure of the active site interface is accompanied by a distortion of helix 1 in the C-terminal domain of the α-subunits at Gly 388 (Fig. 3*c*). The preference of β_{TP} and β_{DP} for AMP-PNP and ADP, respectively, is probably caused by small conformational differences near to their P-loops, the most significant being the shift of α_{DP}-Arg 373 by 1.5 Å towards the β-phosphate moiety in β_{DP}.

Rotational catalysis

The asymmetry required by the binding change mechanism of ATP synthesis (Fig. 3*d*), particularly in the region of the active sites, is evident in the crystal structure. The three different conformations of the subunits β_E, β_{TP} and β_{DP} could be identified on the structural grounds described above as representing the open, loose and tight states, respectively. This asymmetry is associated with the asymmetric positioning of the γ-subunit relative to the $\alpha_3\beta_3$ subassembly.

The nucleotides that are found in the catalytic sites in the crystals appear at first sight to be inconsistent with this proposal, as the tight site (β_{DP}) should be occupied by ATP, and the loose site (β_{TP}) by ADP or ADP plus phosphate. However, if, as we propose, the structure in the crystal is that of the ADP-inhibited form of the enzyme, ADP (but not P_i) would be bound at the tight catalytic site. The presence of AMP-PNP in the loose site may be due to its 50-fold higher concentration over ADP, and to the absence of P_i from the crystallization medium. Alternatively, it is possible that the sequence of events during catalysis differs from that shown in Fig. 1, and that the 'tight' site of newly synthesized ATP first becomes 'loose' and then 'open'. Crystals of F_1 grown in the presence of P_i and alternative nucleotides (ATP, ATP-γS) could help to clarify this ambiguity.

The transition between the different catalytic states implied by the cycle of binding changes could be achieved by rotation of the catalytic subunits relative to the rest of the ATP synthase. This hypothesis cannot be tested by a single 'snapshot' of F_1-ATPase provided by the crystal structure, but several features of the structure are compatible with rotation of the $\alpha_3\beta_3$ subassembly relative to the γ-subunit (and presumably the remainder of the intact synthase). First, the sleeve surrounding the C terminus of the γ-subunit looks like a molecular bearing, lubricated by a hydrophobic interface (Fig. 5*a* and *b*). The crown of β-barrels above it, which form a continuous β-sheet linking all six subunits, holds the structure together, and the top of the nucleotide domain also contributes to its stability. Second, the central cavity would permit passage of the long N-terminal helix of the γ-subunit within the core of F_1 during rotation, with minimal contact between the γ-subunit and the rest of the interior of the particle (Fig. 5*c*). Third, the differences between the three catalytic sites are correlated with the asymmetric position of the γ-subunit. If there is to be an interconversion of the three sites, as the binding-change mechanism suggests, the most obvious way to achieve this is by rotation of $\alpha_3\beta_3$ relative to the γ-subunit. The nature of the interactions between the γ-subunit and $\alpha_3\beta_3$ suggest that the energy barrier opposing rotation is quite small. It could be overcome with energy from conformational changes induced by proton transport and relayed to the active sites via subunit γ and other stalk components. The V_{max} of the bovine ATP synthase is about 400 per second, and twice that value in the chloroplast enzyme[10]. Therefore, the rate of rotation in ATP synthases would be about 130–270 Hz.

Rotation of F_1 in ATP synthase and of the hydrophobic subunits in F_o have been considered in the past[10,35–37]. Attempts to demonstrate rotation have given negative results[10], but the possibility of rotation should be re-examined in the light of the structure. In the F_1-ATPase from *Escherichia coli*, nucleotide-dependent movements of the minor subunits (γ, δ and ε) in the central space between the hexameric arrangement of α- and β-subunits have been observed by cryoelectron microscopy[12] and by changes in the properties of fluorescent labels[38]. In the chloroplast enzyme, there is also evidence of movement of a central mass[39]. The proposed rotation within ATP synthase may be analogous to the rotation of the bacterial flagellum, which is also driven by the transmembrane protonmotive force[40]. A homologue of the β-subunit of ATPase is associated with the flagella complex, and may be involved in its assembly[41]. Assessment of the validity of the analogy between ATPase and flagellae requires appropriate biophysical examination of the ATP synthase based on the structure of F_1, and further detailed structural investigation of both ATP synthase and the flagellar motor. In the ATP synthase, it will be necessary to know more about the coupling mechanism (of which the γ-subunit and probably δ and ε are part) linking the catalytic sites with proton transport in the F_0 domain. The characterization of the proteins in the bovine stalk[16,42] and the *in vitro* assembly of an F_1/stalk complex[42] are necessary steps towards that goal. Little is known about the structure of the intrinsic membrane domain, F_o, and determining its high-resolution structure and understanding the structural basis of the coupling of proton transport to ATP synthesis are still challenges for the future. □

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